Solvency Field Theory V13.3

The First Locally Real Atom:
From Energy at c to Hydrogen

Narrative companion to the
V13.3 Atom Assembly

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Energy at c closes into particles. Particles settle geometry. Settlement at nuclear compactness binds quarks. Settlement at atomic compactness binds electrons. One stuff, one mechanism, every scale from the Compton radius to the Bohr radius.

Publication boundary. V13.3 assembles the first locally real atom at leading nonrelativistic order, with α as input and the proton interior treated as an effective guarded source. It does not derive α, the proton's internal structure, fine/hyperfine splitting, the Lamb shift, or molecular chemistry. It builds hydrogen from settlement mechanics and gets the right answer.

Contents

1 Eight species build the universe

The Standard Model catalogs 61 named particles. SFT sorts them into three tiers by persistence.

Tier 1 — Persistent builders: electron, up quark, down quark, three neutrinos, photon. Eight species (plus antiparticles). These build 100% of the visible universe. Every atom. Every star. Every person.

Tier 2 — Unstable visitors: muon, tau, strange, charm, bottom, top, W, Z, Higgs, gluon. Ten transient species. Real but short-lived. They appear in collisions and decay into Tier 1 particles. None persist. None constitute matter.

Tier 3 — Virtual/hypothetical: graviton, virtual particles, dark matter candidates, supersymmetric partners. Zero confirmed. Mathematical infrastructure or undetected proposals.

V13.3 uses only Tier 1 species to build the atom. Everything else is either a brief settlement event (Tier 2) or unnecessary (Tier 3).

2 The stability band

The persistent charged matter universe spans a remarkably narrow mass range:

Particle	Mass (MeV)	r_{spin} (fm)
Electron	0.511	193.1
Up quark	2.2	44.9
Down quark	4.7	21.0

A factor of 9.2 in mass. Less than one order of magnitude. That is the ENTIRE basis of visible matter.

In SFT, these are the ground-state closure modes for their respective charge/spin families. Each one is the MINIMUM VIABLE COMPACTNESS for its quantum numbers — the lightest self-consistent orbit that can carry that specific charge and spin.

The heavier fermions — muon, tau, strange, charm, bottom, top — are higher closure modes in the same families. Valid orbits. Self-consistent geometry. But they decay because a lighter mode with the same quantum numbers exists. Water flows to the lowest available crater. Energy flows to the lightest available closure.

3 Inside the proton

Two up quarks and one down quark. Total charge +1. Total current-mass seed: $2 \times 2.2 + 4.7 = 9.1\,\text{MeV}$.

The proton weighs 938.27 MeV.

The quarks contribute 1% of the proton's mass. The other 99% is settlement field energy between them. The settlement geometry of three closed orbits packed into a femtometer-scale volume contains 100 times more energy than the orbits themselves.

Why? Because settlement at nuclear compactness is ENORMOUS. The compactness equation $C_A = A_{\text{write}}E/(\hbar c r)$ evaluated at $r \approx 1\,\text{fm}$ with the proton's full mass is still only $\sim 10^{-40}$ in raw gravitational terms. But the L_{eff}-compounded settlement at nuclear distances amplifies the effective binding by the same mechanism that produces the mass hierarchy. The settlement COMPOUNDS within the nuclear volume.

The required settlement tension — roughly 0.93 GeV/fm across the proton's diameter — is nuclear-natural: about 4.7 $\hbar c$/fm total, or 1.57 $\hbar c$/fm per quark-quark link. Not exotic. Not absurd. Order unity in the natural units of nuclear physics.

Comparing to the pion (the simplest quark-antiquark bound state at $\sim 140\,\mathrm{MeV}$), the proton needs a three-body enhancement factor $\eta_3 \approx 2.2$–2.4 beyond three independent pion-like links. In $\mathrm{L_{eff}}$ terms, this extra factor corresponds to only $\Delta u \approx 0.05$ of additional compounding. The baryon junction is a small perturbation on the pion-scale settlement link.

V13.3 does not derive the proton's internal structure. It treats the proton as a guarded effective source: charge +1, mass 938 MeV, interior kernel open. This is justified because the atom operates at a completely different scale.

4 The Coulomb well

The proton is a net +1 directional settlement source. Its directional settlement — the circulation-direction component that carries charge information — extends outward into three-dimensional space.

Conserved directional flux through a sphere of area $4\pi r^2$ gives:

$$|\mathbf{D}(r)| \propto \frac{1}{r^2}, \qquad V(r) = -\frac{\alpha\hbar c}{r}. \tag{1}$$

The $1/r^2$ field and $1/r$ potential are as inevitable as $1/r^2$ gravity — both come from the Green kernel of three-dimensional space. The difference is that gravity is the scalar (direction-blind) settlement component, while electromagnetism is the directional (circulation-sensitive) component.

The electron, with charge -1 (opposite circulation), finds this well attractive. Same-direction circulation repels. Opposite-direction attracts. The electron falls toward the proton.

At the Bohr radius, the proton's physical radius ($r_p \approx 0.84\,\mathrm{fm}$) is 1.6×10^{-5} times smaller than the electron's orbital distance ($a_0 \approx 52{,}917\,\mathrm{fm}$). The proton is effectively a point source. Its internal structure is invisible to the electron.

5 Why the electron doesn't fall in

The directional settlement pulls the electron toward the proton. What stops it?

The electron is a closed orbit of energy at c. It has a definite spatial extent — its closure support domain. Compressing the electron into a smaller region costs energy because the closure has to be maintained against the compression. The tighter you confine the orbit, the faster the internal momentum, the higher the kinetic settlement cost.

This is the uncertainty principle stated mechanically. $\Delta p \approx \hbar/r$ gives kinetic energy:

$$K(r) = \frac{\hbar^2}{2\mu\, r^2}. \tag{2}$$

The total energy at orbital distance r is:

$$E(r) = \underbrace{\frac{\hbar^2}{2\mu\, r^2}}_{\text{closure maintenance}} - \underbrace{\frac{\alpha\hbar c}{r}}_{\text{directional settlement}}. \tag{3}$$

The first term pushes out. The second term pulls in. At large r, the $1/r$ attraction dominates — the electron falls inward. At small r, the $1/r^2$ maintenance cost dominates — compressing the electron costs more than the attraction gains.

The minimum — the equilibrium — is at:

$$\boxed{a_0 = \frac{\hbar}{\mu\, c\, \alpha} \approx 52.95 \text{ pm.}} \tag{4}$$

The binding energy at this radius:

$$\boxed{E_{\text{bind}} = \frac{\mu\, c^2 \alpha^2}{2} \approx 13.6 \text{ eV.}} \tag{5}$$

The electron sits 137 times farther out than its own Compton wavelength. The electromagnetic coupling $\alpha \approx 1/137$ is too weak to compress the electron down to its own closure scale. The electron barely notices it's bound — its binding energy is 0.0027% of its rest mass.

6 Hydrogen

The proton provides the well. The electron provides the occupant. The settlement mechanics provide the binding. The result is hydrogen:

$$p^+ + e^- \to H^0. \tag{6}$$

Charge: $+1 + (-1) = 0$. Neutral.

Mass: $938.27 + 0.511 - 0.0000136 \approx 938.78\,\text{MeV}$.

Bohr radius: $52.95\,\text{pm}$ (infinite-proton) or $52.95\,\text{pm}$ (reduced-mass).

Binding energy: $13.6\,\text{eV}$.

The atom has three layers of settlement operating at three completely different compactness scales:

Nuclear ($r \approx 1\,\text{fm}$): Three quarks bound by compounded settlement. 929 MeV of field energy. 99% of the proton's mass is settlement geometry. This is the strong force.

Electromagnetic ($r \approx 53\,\text{pm}$): Electron bound by directional settlement. 13.6 eV binding. 0.0027% of the electron's rest mass. This is the electromagnetic force.

Gravitational ($r \approx$ stellar): The atom's total mass contributes to the scalar settlement field through $C_A = A_{\text{write}} E/(\hbar c r)$. Negligible at atomic scales. Dominant at stellar scales. This is gravity.

One settlement mechanism. Three regimes. Three "forces." All the same thing at different compactness scales.

7 The assembly chain

Starting from nothing but energy at c and no-overwrite:

V13.0: Energy closes into particles. Nine modes from 3+1 slots and no-overwrite. Mass from $L_{\text{eff}} = 3\pi - \frac{1}{2}\ln(25/13)$. The stability band selects three persistent charged modes.

V13.1: Particle settlement accumulates. Compactness gradients form. $G = A_{\text{write}} c^3/\hbar$. Things fall toward the slowest clock.

V13.2: The same writes demand boundary capacity. $H = \dot{N}/I$. The universe runs at break-even. One Hubble time of reserve.
V13.3: Settlement at nuclear compactness binds quarks into a proton. Settlement at atomic compactness binds an electron around the proton. Hydrogen forms.

From energy to atom in four documents. From the Compton radius to the Bohr radius through one mechanism. From the particle that creates time to the atom that creates chemistry.

8 What comes next

Hydrogen is the first atom. Not the last. The road from here to the periodic table, to molecules, to chemistry, to biology requires:

α derived from the directional-to-scalar settlement ratio. This pins every electromagnetic prediction to a derived constant rather than an input.

Neutrons assembled from *udd*. Similar to the proton but with charge 0 and slightly higher mass. Nuclear stability requires understanding why *udd* is unstable as a free particle but stable inside nuclei.

Nuclei assembled from protons and neutrons. Nuclear binding energy. The semi-empirical mass formula. Why iron is the most stable nucleus.

Multi-electron atoms. Electron shell structure. The Pauli exclusion principle from no-overwrite (each electron needs a unique settlement address). The periodic table.

Molecules from shared settlement between atoms. Chemical bonds. Molecular geometry.

Each step uses the same settlement mechanics operating at the same or slightly different compactness scales. No new forces. No new particles. No new principles. Just the same no-overwrite constraint producing richer structure at each level of assembly.

9 The sentence

V13.3 can be stated in one sentence:

> *Three packets of energy at c, orbiting at nuclear compactness, produce a proton; one more packet, orbiting at atomic compactness around the first three, produces hydrogen; and the settlement mechanics that bind them are the same mechanics that curve spacetime, expand the universe, and create time.*

One stuff. One rule. One atom.

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References

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